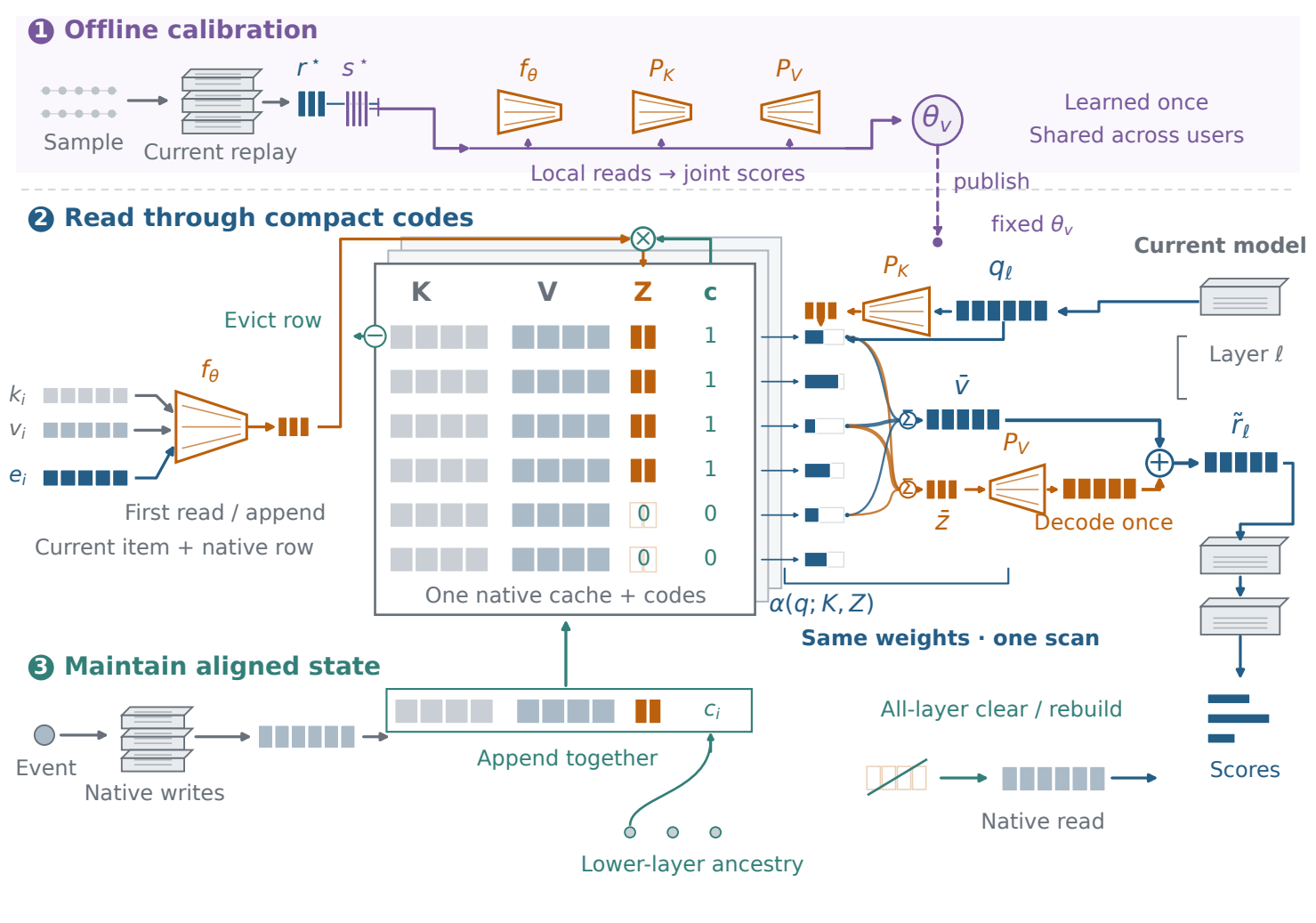
1 Offline calibration
Sample
Current replay
$r^\star$
$s^\star$
f_θ
P_K
P_V
θ_v
Learned once
Shared across users
Local reads → joint scores
publish
fixed θ_v
2 Read through compact codes
Current model
Evict row
K
V
Z
c
1
1
1
1
0
0
0
0
One native cache + codes
P_K
q_ℓ
Layer ℓ
k_i
v_i
e_i
f_θ
First read / append
Current item + native row
$\bar{v}$
$\bar{z}$
P_V
$\tilde{r}_\ell$
Decode once
$\alpha(q; K, Z)$
Same weights · one scan
3 Maintain aligned state
Event
Native writes
c_i
Append together
Lower-layer ancestry
All-layer clear / rebuild
Native read
Scores